LIG: Layer-wise Integrated Gradients for Within-Layer Flow Analysis in Transformers

MON-19

Poster (PDF) · My page

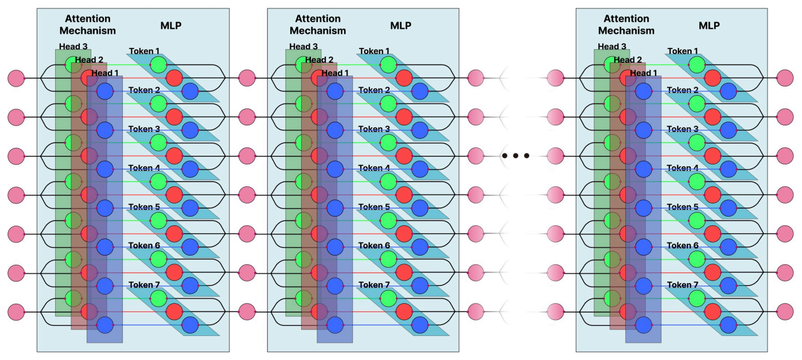
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Attributing ATT and MLP separately and composing them matches whole-layer IG best when their baselines follow the whole-layer baseline.

LIG (Layer-wise Integrated Gradients) applies Integrated Gradients (IG) at the Multi-Head Attention (ATT) and multilayer perceptron (MLP) boundaries and composes them. Best on BERT-base, 1,700 Penn Treebank (PTB) sentences: whole-layer ITB → ITB for ATT, ATTITBa=0 for MLP; whole-layer Zero → Zero for both.

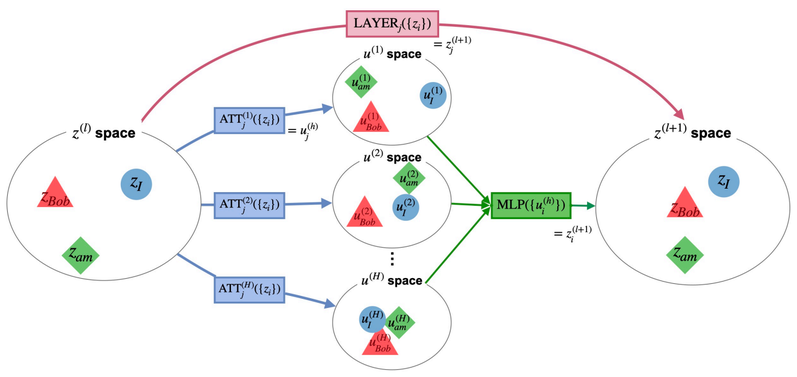
Project page
https://eightsuzuki.github.io/layer-wise-integrated-gradients/

1 What flows inside a layer?



Each layer as a **dynamic graph**: tokens z_i (pink) and per-head attention outputs $u_i^{(h)}$ are nodes; information crosses the ATT and MLP boundaries.

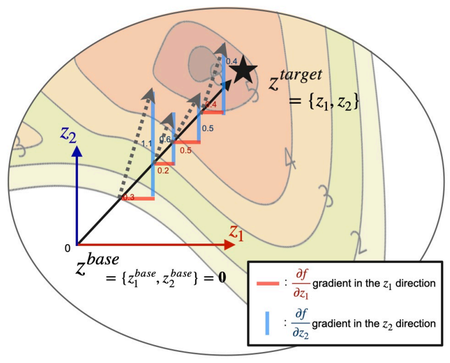
We attribute each boundary with IG (bottom path) and ask whether the composition matches IG on the whole layer (top path).



Method	Attributes	Limitation / LIG's gain
PRIOR Attention map [7]	weights $\alpha_{i,j}$	ignores values and MLP
PRIOR Rollout [5]	attention across layers	ignores values and MLP
PRIOR Norm-based [6]	weighted value norms	no ATT / MLP split
PRIOR AttnLRP [4]	relevance via ATT, MLP	a rule per operation
OURS **LIG**	**IG at each ATT / MLP edge**	**one rule (IG); splits and composes**

Set-to-set IG: IG from each element of a module's *input set* (tokens at ATT, heads at MLP) to each output token. **LIG** applies it at every ATT / MLP boundary and composes the results, as in **L**ayer-wise Relevance Propagation (LRP) [2, 3].

2 IG, and where the path starts



PRIOR IG [1] *line-integrates* the gradient along a path from a **baseline** $z(0)$ to the input z. Any path keeps completeness, so path and start are ours to choose; we use the straight path $z(a) = z(0) + a\,(z - z(0))$:

$$\mathrm{IG}_d = (z_d - z_d(0)) \int_0^1 \frac{\partial f(z(a))}{\partial z_d}\, da$$

f: scalar output (contours); d: input dimension.

Completeness: $\sum_d \mathrm{IG}_d = f(z) - f(z(0))$. The baseline sets **what each contribution is measured against** (§5).

3 LIG: set-to-set IG at each boundary OURS

At each boundary (cells below), apply IG to the input x from its **baseline** $x(0)$ with $f = \|y_j(a) - y_j(0)\|_2$, summing over each input element r (token or head); scores are signed:

$$\mathrm{IG}_{r,j} = \sum_{d \in r} \mathrm{IG}_d, \qquad \widetilde{\mathrm{IG}}_{r,j} = \frac{\mathrm{IG}_{r,j}}{\|y_j(1) - y_j(0)\|_2} \quad \Big(\sum_r \widetilde{\mathrm{IG}}_{r,j} = 1\Big)$$

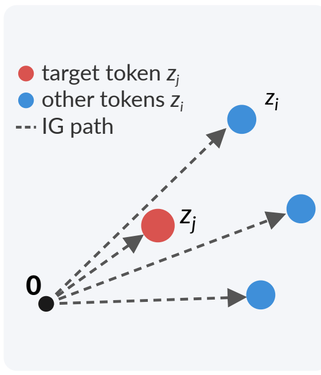
ATT
$z_i \to u_j^{(h)}$: $\mathrm{IG}^{\mathrm{ATT}}_{i,j,h}$

MLP (W_o, LayerNorm, FFN; residual z fixed)
$u_j^{(h)} \to z_j^{(l+1)}$: $\mathrm{IG}^{\mathrm{MLP}}_{h,j}$

Whole layer l
$z_i \to z_j^{(l+1)}$: $\mathrm{IG}^{\mathrm{LAYER}}_{i,j}$

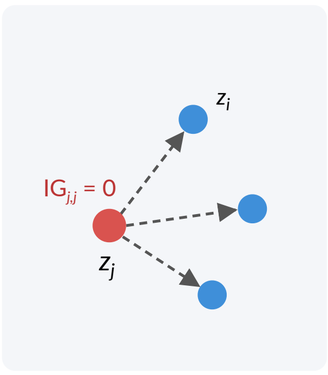
Baselines: how the **input** moves along the IG path, $a = 0 \to 1$ (input-space sketch).

Zero



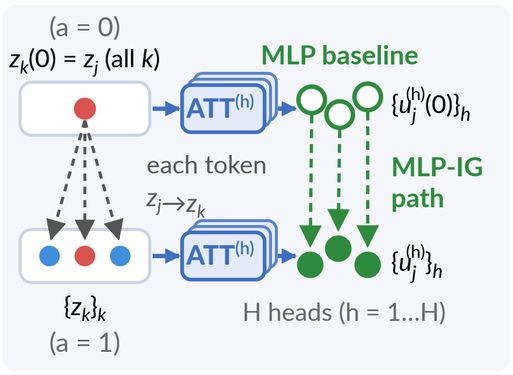
Every path starts at the zero vector $\mathbf{0}$.

Input Token Base (ITB)



All tokens start at the target z_j; z_j itself does not move, so $\mathrm{IG}_{j,j} = 0$.

MLP: ATTITBa=0



At $a = 0$ all tokens are z_j, so each head outputs z_j's value $u_j^{(h)}(0)$: the MLP baseline.

Why ITB? ATT averages token values, so we measure each token *relative to the target* z_j. Its zero self term [8] comes from the reference, not the layer ($\alpha_{j,j} > 0$); two completions estimate it, keeping the total:

$$\mathrm{IG}^{\mathrm{zeroRatio}}_{j,j} = \mathrm{IG}^{\mathrm{zero}}_{j,j} \frac{\sum_{k \neq j} \mathrm{IG}^{\mathrm{ITB}}_{k,j}}{\sum_{k \neq j} \mathrm{IG}^{\mathrm{zero}}_{k,j}}, \qquad \mathrm{IG}^{\mathrm{mapRatio}}_{j,j} \approx \Big(\sum_{k \neq j} \mathrm{IG}^{\mathrm{ITB}}_{k,j}\Big) \frac{\alpha_{j,j}}{1 - \alpha_{j,j}}$$

Why ATTITBa=0? MLP inherits the ATT output at the ITB start ($a = 0$), keeping both modules on one reference.

4 Composing ATT and MLP OURS

Chain the two boundaries, LRP-style, and compare the composed token-level allocation with whole-layer IG:

$$\underbrace{\widetilde{\mathrm{IG}}^{\mathrm{LAYER}}_{i,j}}_{\text{whole layer}} \overset{?}{\approx} \underbrace{\sum_h \widetilde{\mathrm{IG}}^{\mathrm{ATT}}_{i,j,h}\, \widetilde{\mathrm{IG}}^{\mathrm{MLP}}_{h,j}}_{\text{composed}}$$

Not an IG chain rule: the composition omits the residual route, and separately integrated paths need not agree. We measure this *compositional gap*, not faithfulness.

5 Result: does the composition match the layer?

Setup. BERT-base-uncased; PTB dev, 1,700 sentences [9]; $m = 32$ steps. Per sentence, layer l and output token j, the L_2 distance $d_j^{(l)}$ between the two sides of §4 (lower = better).

Mean over (sentence, layer, token); top 3 per block, **bold** = best; compare within blocks.

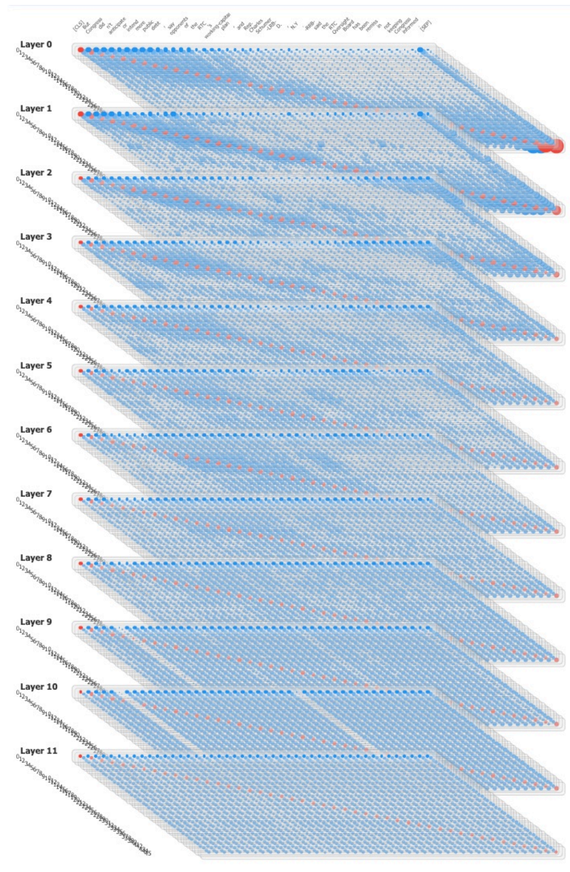
Whole-layer baseline (reference)	Composed: ATT	MLP	L_2 ↓
ITB Input Token Base: all tokens start at z_j	**ITB**	**ATTITBa=0**	**0.305**
	ITB-zeroRatio	ATTITBa=0	0.312
	ITB-mapRatio	ATTITBa=0	0.318
Zero zero vector $\mathbf{0}$	**Zero**	**Zero**	**0.736**
	ITB-zeroRatio	Zero	0.744
	Zero	ATTITBa=0	0.749
LAYER-ITB-zeroRatio whole-layer ITB, self term filled by zeroRatio	**Zero**	**Zero**	**0.735**
	ITB-zeroRatio	Zero	0.742
	Zero	ATTITBa=0	0.748

The best MLP baseline follows the reference: ATTITBa=0 under ITB (for every tested ATT baseline); Zero under Zero and LAYER-ITB-zeroRatio.

6 Use case: reading the flow layer by layer

Whole-layer IG, Zero baseline, layers 0–11, one PTB sentence (qualitative). Columns: input tokens; rows: output tokens; circle size = contribution; red = self, blue = other tokens.

Layer 0: mass on each token and its neighbors; **layers 9–11:** spread across the sentence.



7 Scope and next steps

- **Limits:** BERT-base / PTB only; straight paths may leave the data distribution; ITB completion uniqueness is open; §6 is qualitative; L_2 discards direction. Needing only module boundaries and gradients, LIG can extend to other architectures (untested).
- **Next:** decoder-only models; comparison with EAP / EAP-IG [10, 11]; intervention-based checks.

[1] Sundararajan+ ICML 2017 · [2] Bach+ PLOS ONE 2015 · [3] Montavon+ LRP overview 2019 · [4] Achtibat+ (AttnLRP) ICML 2024 · [5] Abnar & Zuidema ACL 2020 · [6] Kobayashi+ EMNLP 2020 · [7] Jain & Wallace NAACL 2019 · [8] Sturmfels+ Distill 2020 · [9] Marcus+ Treebank-3, LDC 1999 · [10] Syed+ BlackboxNLP 2024 · [11] Hanna+ COLM 2024.
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